

ED_Project_Proposal Standard Deviants

Ndumiso, Nyawo and 26396769 Lebohang, Maleke, 28293436 Masiu, Mohoang, 28298918
Rudairo, Nyamande, 28023366 Lwazi, Zwane, 28913884

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AIM. To investigate whether a student's faculty of study affects the time required to solve a standardised logical reasoning test, comparing mean completion times across faculties using one-way ANOVA.

RESEARCH QUESTION. Does faculty of study influence the average time taken by Stellenbosch University students to solve a standardised logical reasoning test?

Statistical Hypotheses

Means model: $H_0 : \mu_1 = \mu_2 = \mu_3$ (population mean completion time is equal across Engineering, Science, and EMS)
 $H_1 : \text{at least one } \mu_i \neq \mu_j, \text{ for } i \neq j$ (at least one faculty's mean differs)

Equivalent general linear model: Each observation (a student's completion time) is modelled as the sum of three components: the grand mean, a faculty-specific effect, and a random error term. The random error term is assumed to be normally distributed with mean zero and constant variance.

Experimental Design

Experimental units: Individual undergraduate students enrolled at Stellenbosch University. **Factor:** Faculty of study. **Treatment levels:** Engineering, Science, EMS. **Response variable:** Total time (seconds) to complete the reasoning test. **Replication:** 10 students per faculty (n = 30 total).

Experimental Procedure

Each participant completes the identical logical reasoning test under identical conditions. Timing starts when the test begins and stops once the final answer is submitted; time (seconds) is recorded as the response. Participants work individually without calculators, phones, AI tools, or internet access.

Controlled Variables

1. **Test instrument** – identical items, order, and difficulty for all participants.
2. **Academic level** – participants restricted to a similar year of study.
3. **Administration** – questions typed for consistency.
4. **Time of day** – sessions held at similar times to control fatigue/alertness.
5. **Resources** – no assistance, calculators, phones, AI, or internet permitted.

Randomisation

1. Students within each faculty are randomly selected from consenting volunteers to minimise selection bias.
2. Testing order is randomised across faculties (not block-tested by faculty) to prevent time-of-day confounding.
3. All participants complete the identical assessment under identical conditions.

Ethical Considerations

Participation is voluntary, with informed consent covering purpose, procedure, duration, and right to withdraw (consenting participants are recorded on video). The test poses minimal risk; participants may withdraw if uncomfortable. Equal, identical conditions are applied across faculties for fair treatment. No names or student numbers are recorded, ensuring anonymity and confidentiality.

Planned Statistical Analysis

Data will be summarised descriptively (mean, median, SD, min, max) with boxplots comparing faculties. A one-way ANOVA will test for differences in mean completion time between faculties, with normality and homogeneity of variance.